

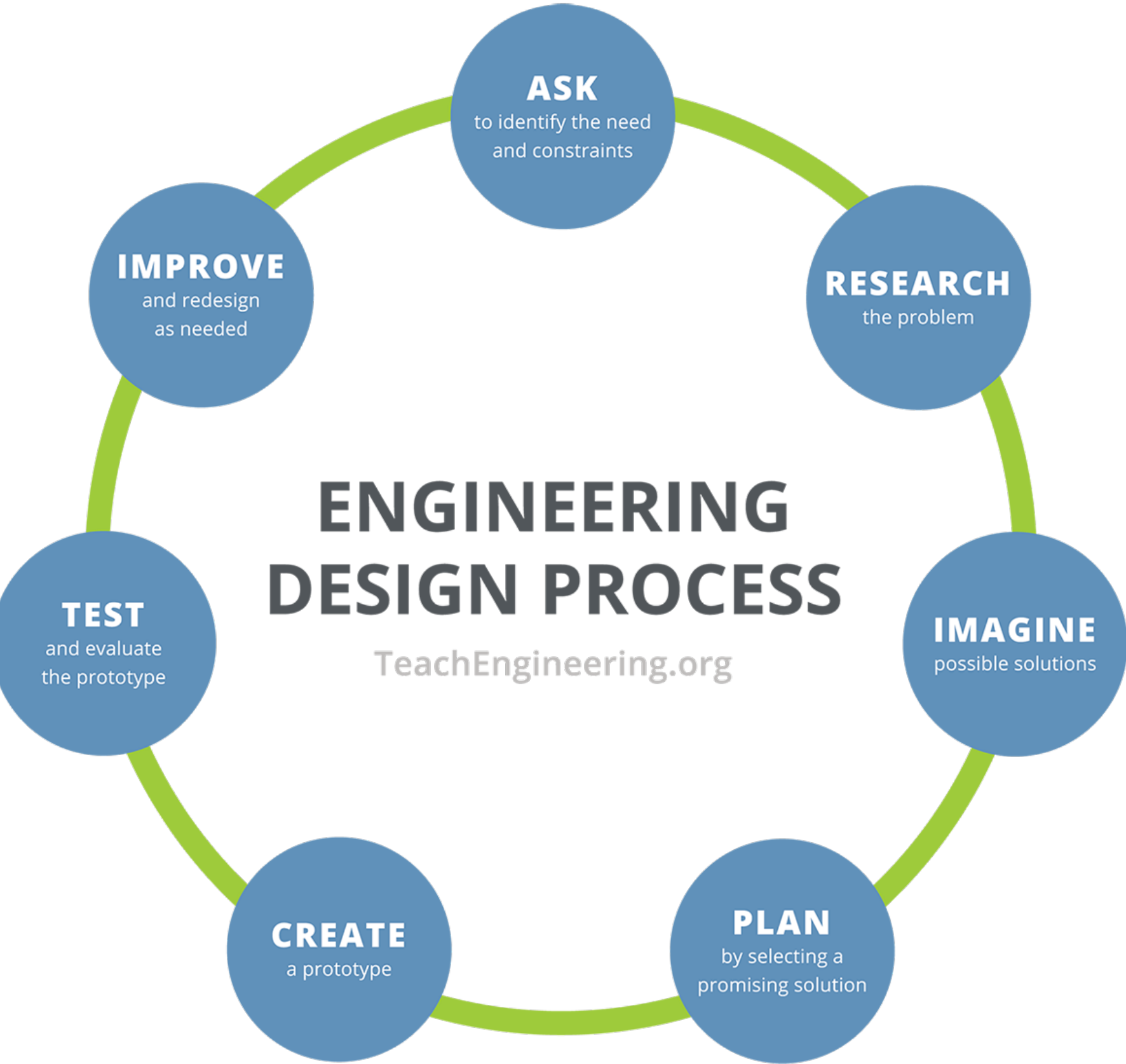
Abstract

This project aims to design and develop a fixed-wing radio control airplane optimized for lift and power efficiency while maintaining a short take-off distance. The primary objective is to create an aircraft capable of carrying a 2-kilogram payload with a minimized empty weight. The design process will focus on achieving optimal aerodynamic performance, structural integrity, and maneuverability . This project serves as a preparation for participation in the SAE Aero Design Competition, where the designed aircraft will be put to the test against other university competing teams.

Overall Design



Figure 1. Enter Caption



Wing Design

Lift

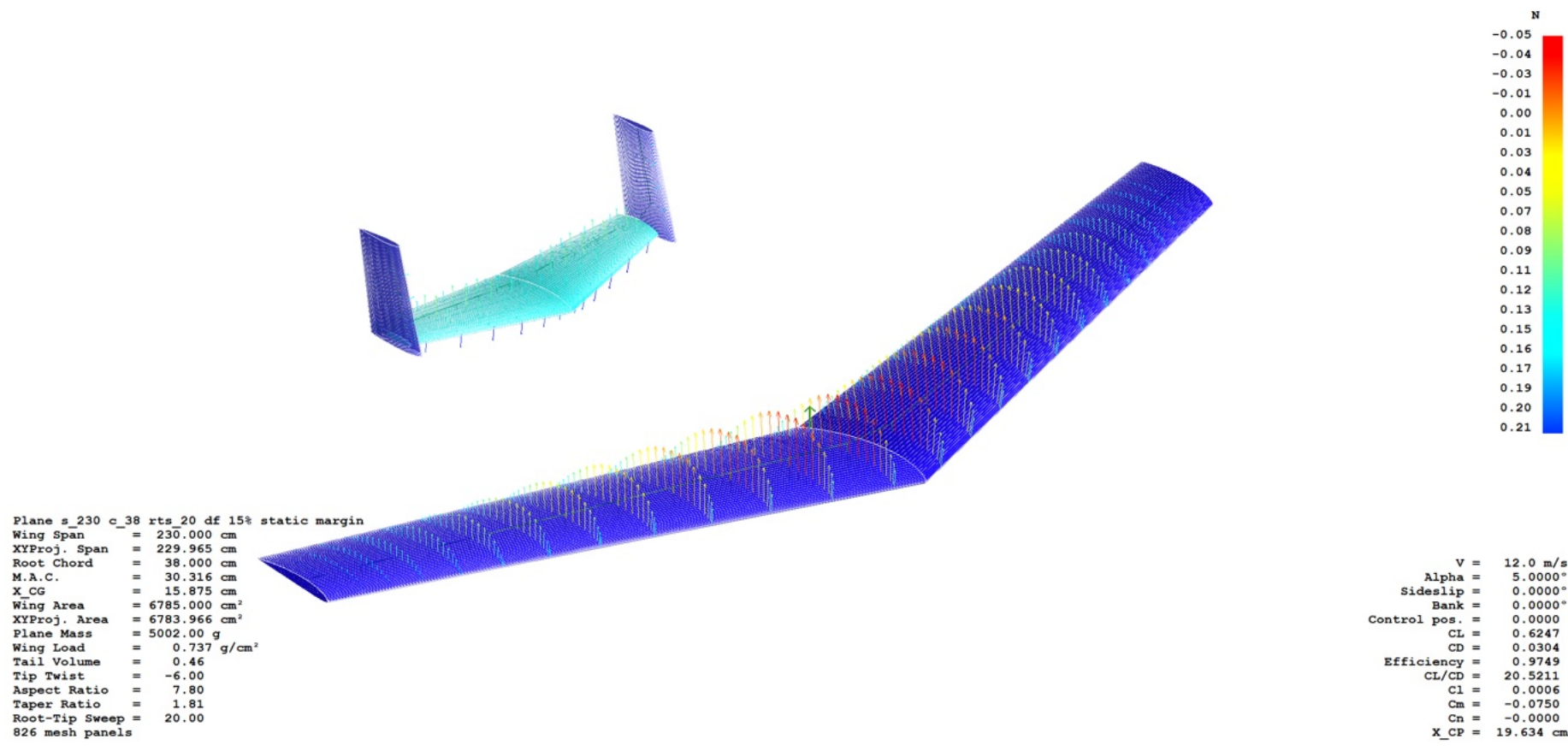


Figure 2. Lift generated by wing

Drag

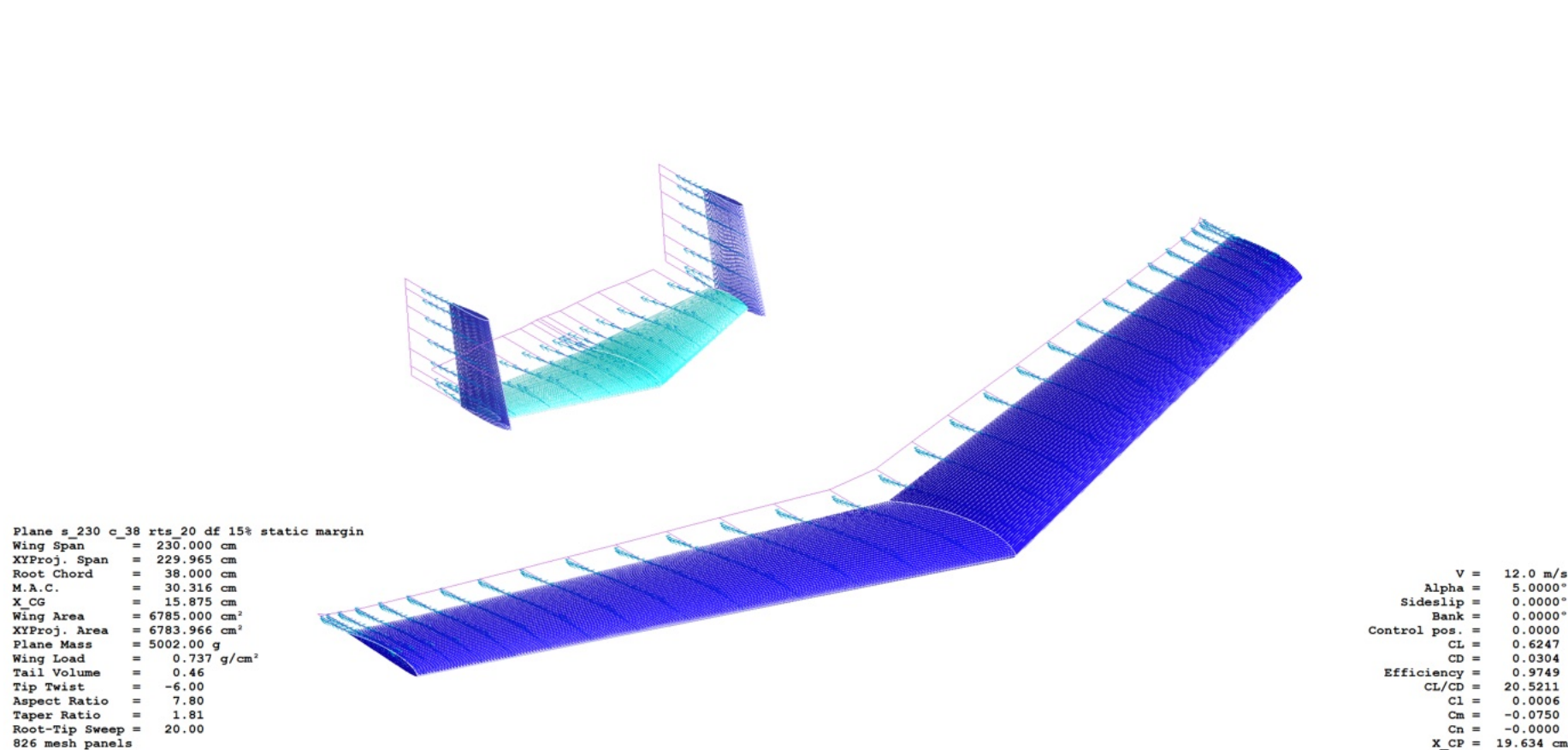


Figure 3. Lift-induced drag

Stability Analysis

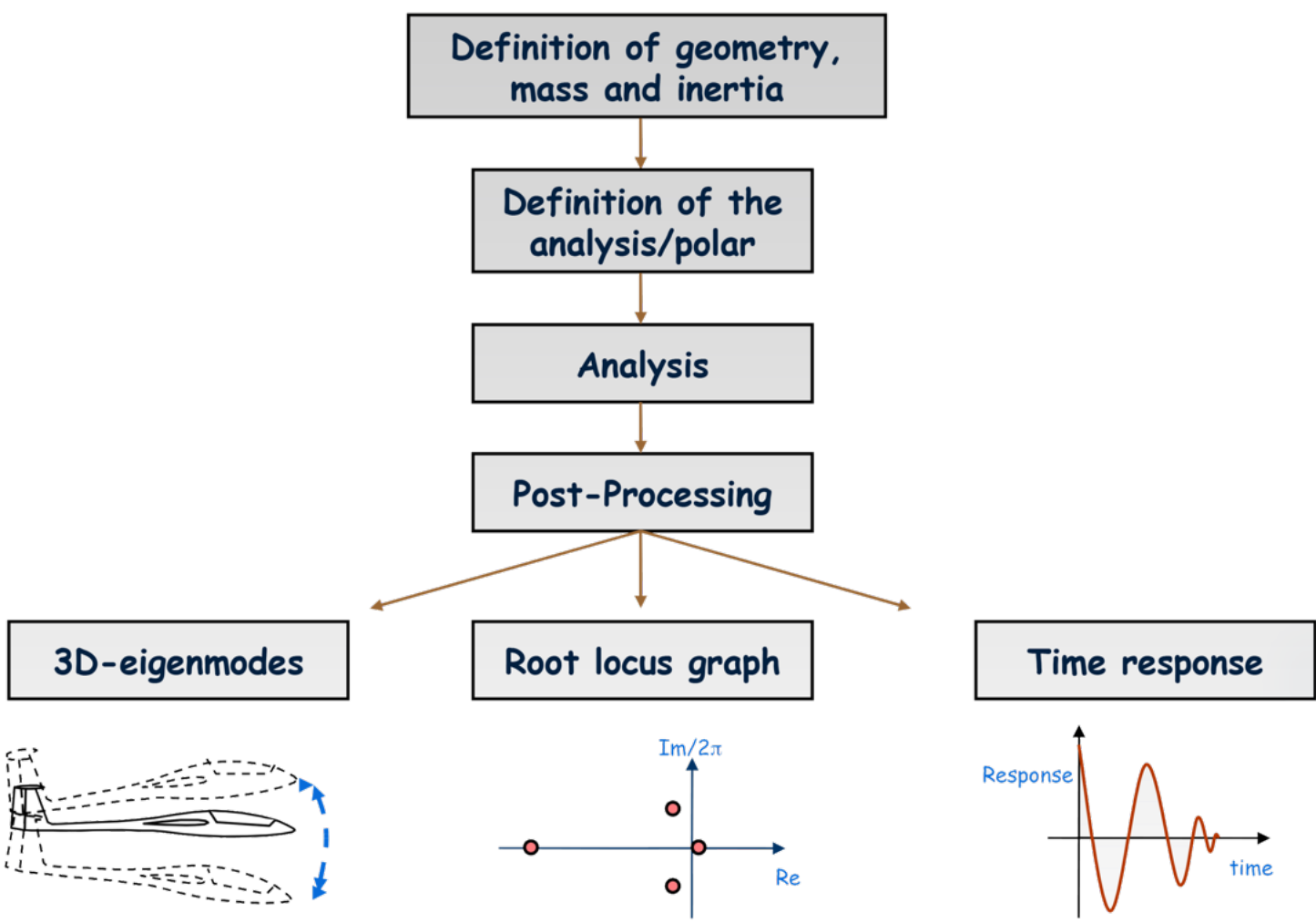


Figure 4. Steps for stability analysis

Propulsion, Electronics, and Control

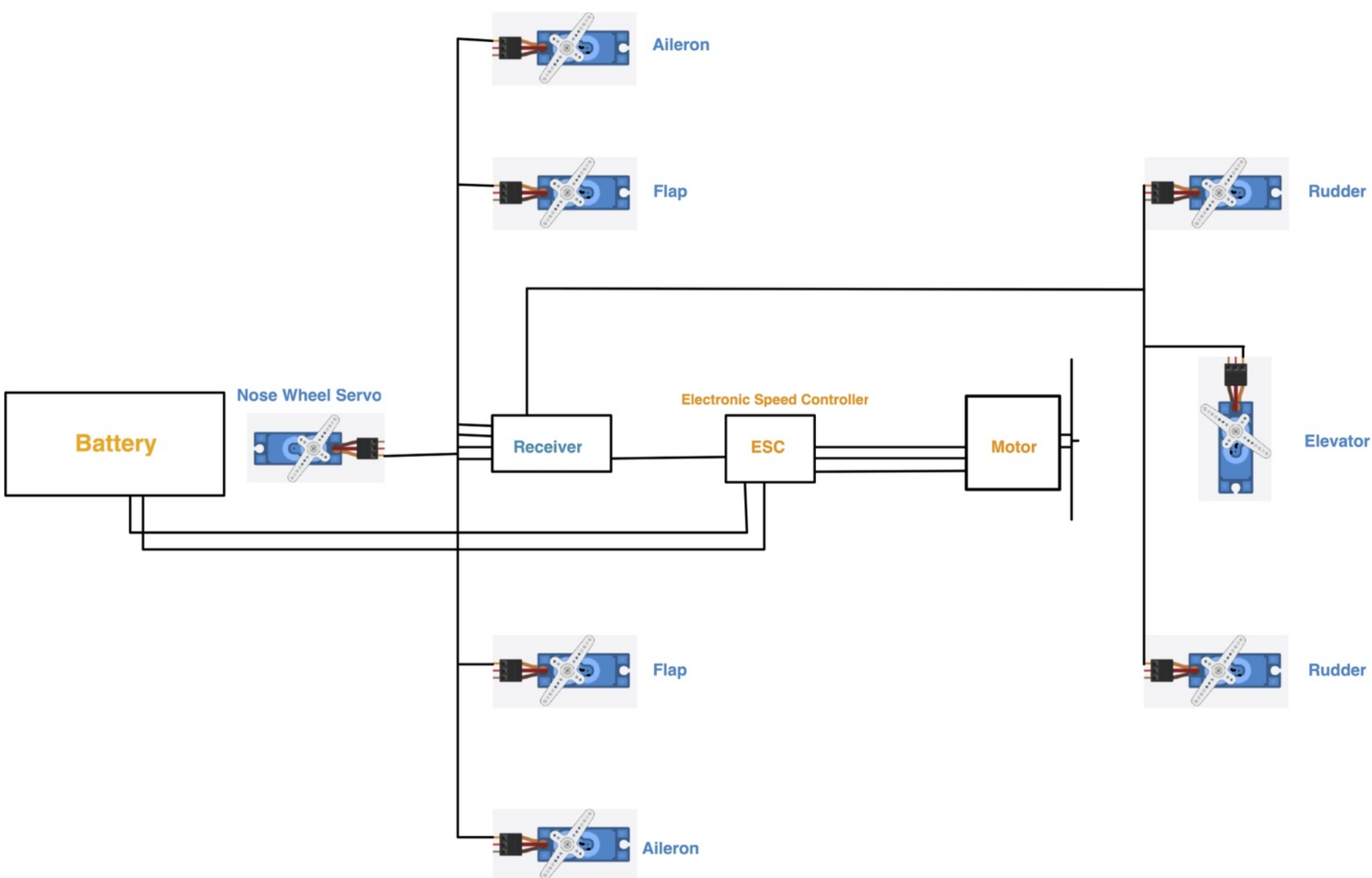


Figure 5. Wiring diagram

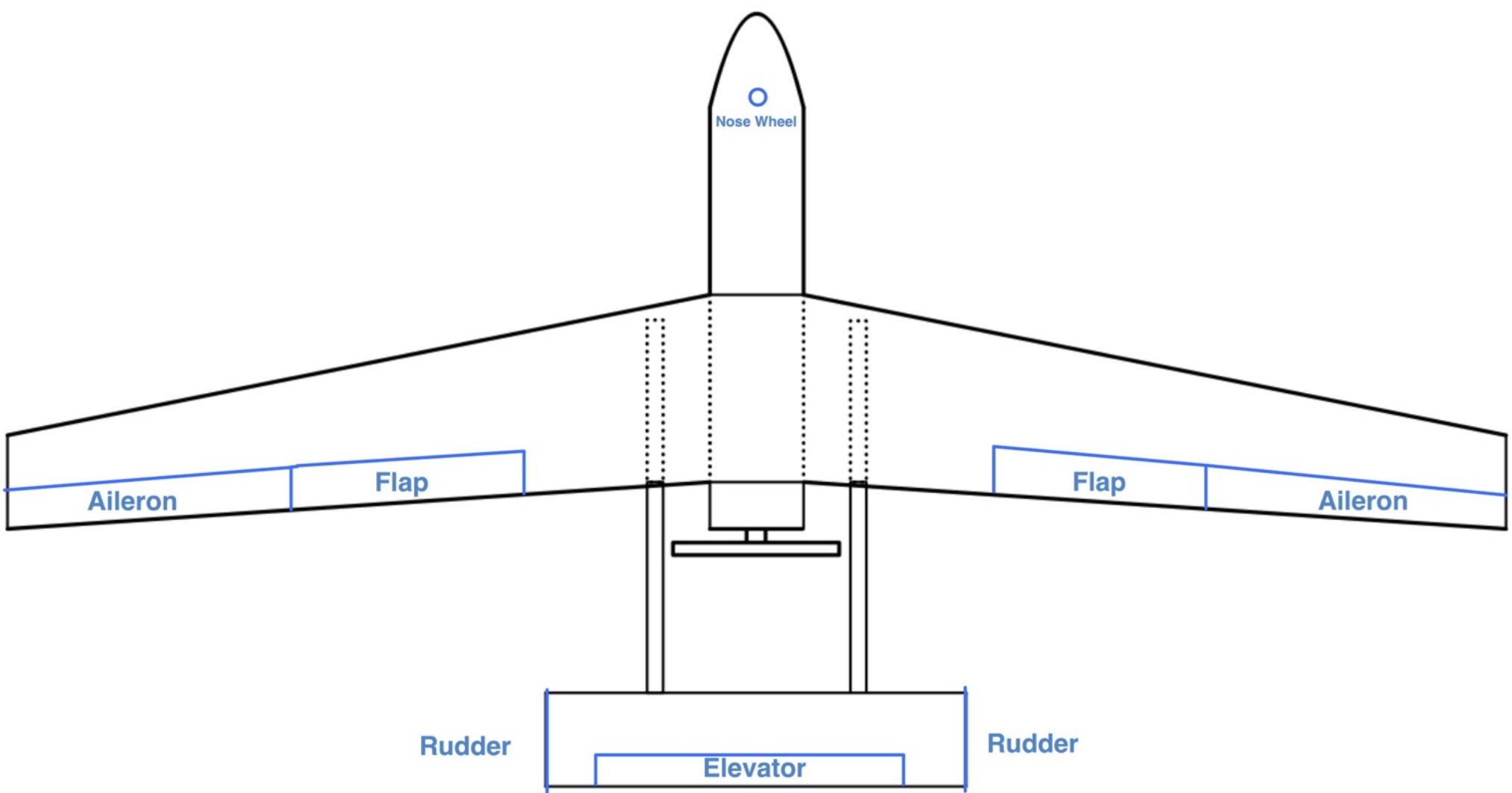


Figure 6. Control

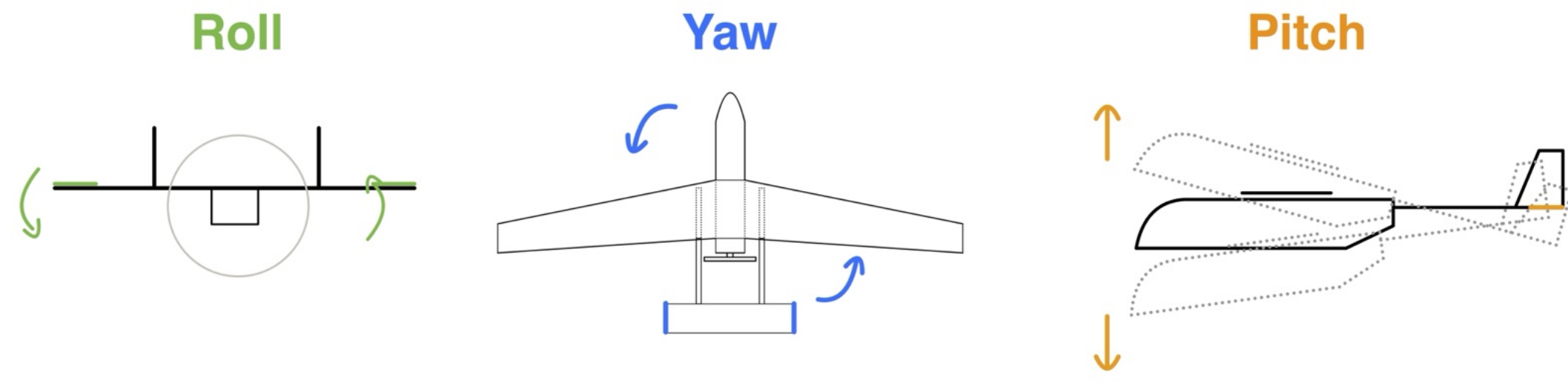


Figure 7. Control

CAD Modeling and Mechanical Design

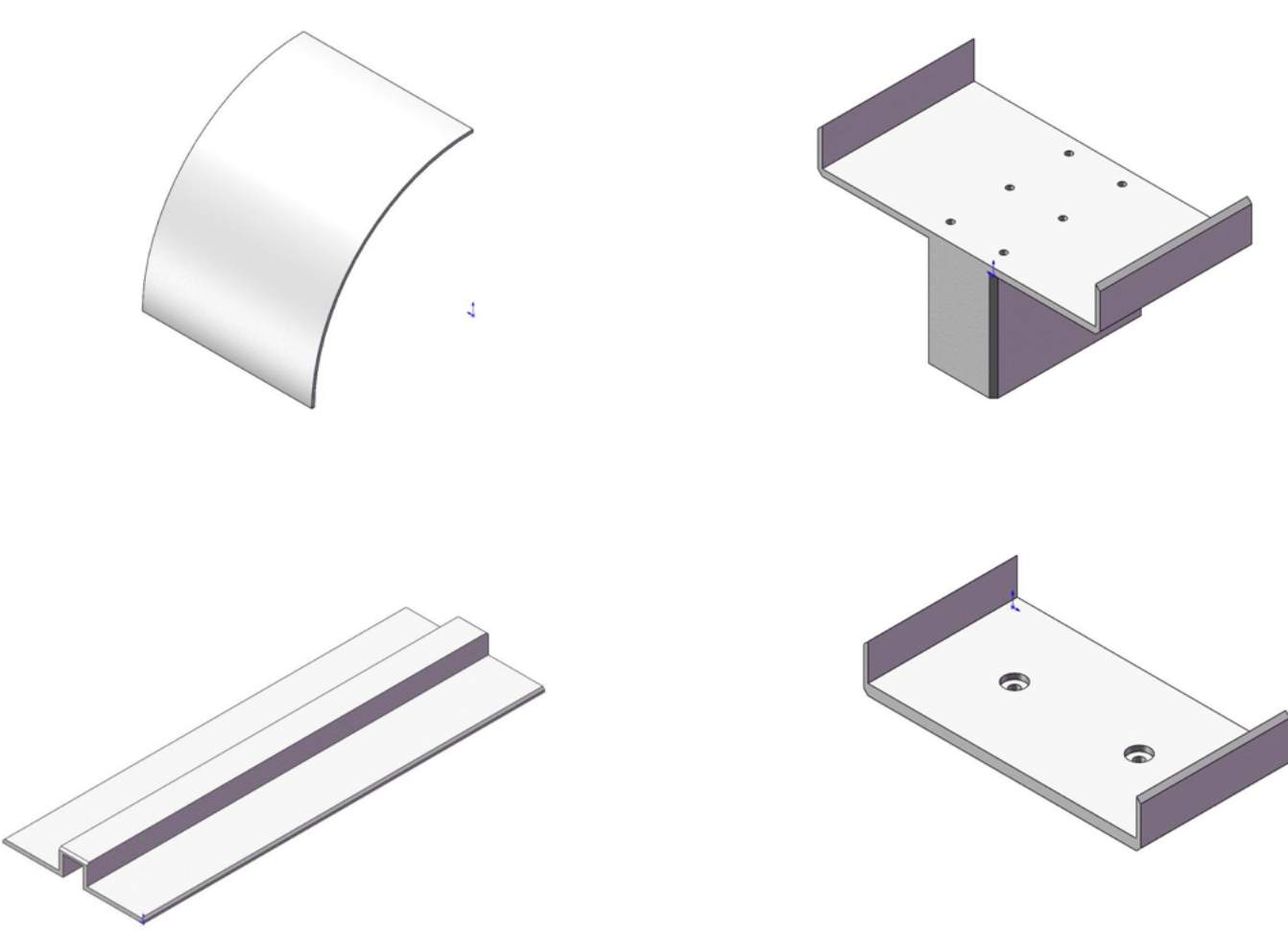


Figure 8. Models of 3D printed landing gear connectors, empennage connectors, and faceplate.

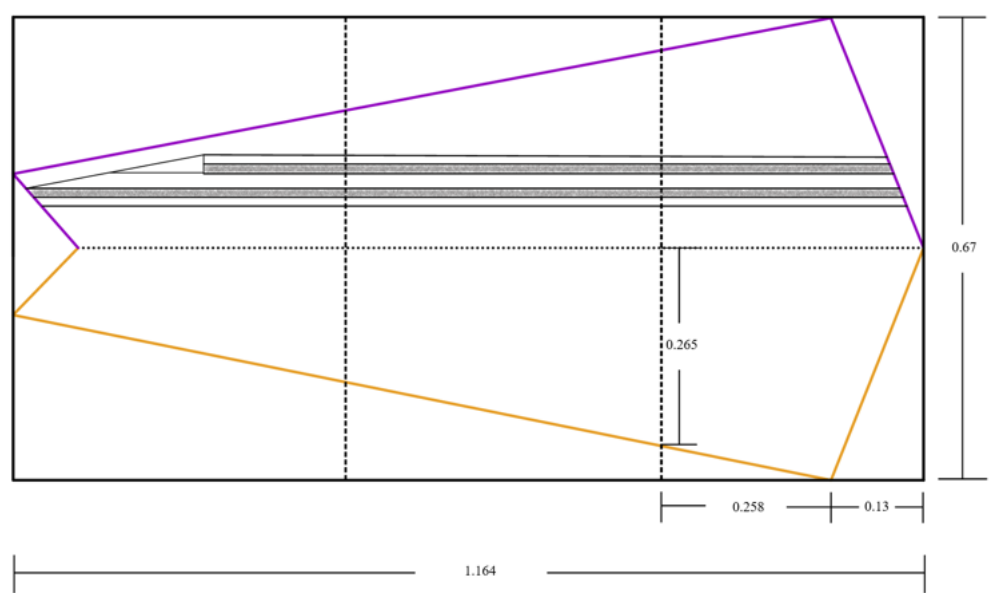


Figure 9. Structure of a foam wing

Improvement and Future Development

Adding a Rudder

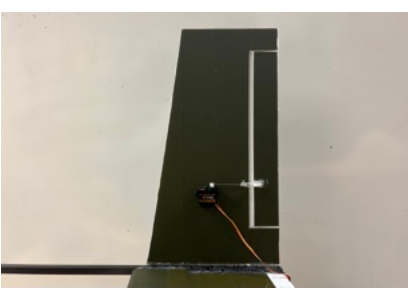


Figure 10. In the initial construction phase, the rudder was not installed, as yaw control can be achieved through the coordinated use of the ailerons and elevator.

Tilted Self-stable wing



Figure 11. Tilted wing Design